

Gene Pulser® Electroporation

Cell Type Other Cell Types

Molecules Electroporated DNA: plasmid constructs (containing VSG- gene, promoter *T. brucei*)

Species Used *Trypanosoma brucei brucei*, AnTat 1.3A, (blood- stream forms); EATRO 1125 (procyclic forms).

Before the Pulse

Cell Growth Medium Cunningham's medium + 15% Fetal Calf Serum
(See references in notes)

Growth Phase at Harvest Mid- log phase

Pre-pulse Incubation Not given

Wash Solution Not given

The Pulse

Instruments Used Gene Pulser® apparatus

Electroporation Temperature Ambient temperature

Electroporation Medium Zimmerman's post-fusion medium

Cuvette Gap 0.4 cm

Cell Density 2 x 10⁷ / ml

Voltage 1.5 kV

Volume of Cells 500 µl

Field Strength 3.75 kV/cm

DNA Concentration 1 mg / ml

DNA Resuspension Buffer TES buffer

Capacitor 25 µF

Volume of DNA 20 to 50 µl

Resistor (Pulse Controller) none Ω NOT

After the Pulse

Time Constant Not given

Outgrowth Medium Cunningham's medium/Baltes Medium (procyclics) / (Bloodstream forms)

Relevant Publications and/or Comments

Note: exponential values designated in parentheses.

Outgrowth Temperature 27 °C procyclics/ 37° C bloodstream forms

Length of Incubation 12 to 18 hours

Selection Method or Assay Used CAT assay

The conditions described are not those used in my own publication, Jefferies, *et. al.*, 1991 *Mol. Cell. Biol.* **11**: 338-341, but were used by Clayton *et. al.*, 1990 *Mol. Cell. Biol.* **10**: 3036-3047, to transfect procyclic trypanosomes.

Electroporation Efficiency Not done

**It is NOT RECOMMENDED to use high voltage with out the Pulse Controller.

Per Cent Survival 30 to 60%

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Survey Number

191